HDOC DAY-56

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BATCH-15

Question

Write a C program to find the first negative integer in every contiguous subarray of size k. Display 0 if no negative integer exists and -1 if k is invalid.

Aim

To write a C program to find the first negative integer in every sliding window of size k using a variable-length array, pointers, functions, and nested loops.

Algorithm

1. Start.
2. Read the size n of the array.
3. Declare a variable-length array arr[n].
4. Read the array elements and value of k.
5. Pass the array to first_negative() using a pointer.
6. Check whether k is valid. If not, print -1.
7. Use an outer loop to traverse each window.
8. Use an inner loop to find the first negative integer.
9. Print the negative integer, or 0 if none exists.
10. Stop.

>C PROGRAM;

~/day56.c - Mousepad

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```c
#include <stdio.h>

void window_maximum(int *arr, int n, int k) {
    if (k <= 0 || k > n) {
        printf("-1");
        return;
    }

    for (int i = 0; i <= n - k; i++) {
        int max = *(arr + i);

        for (int j = 1; j < k; j++) {
            if (*(arr + i + j) > max) {
                max = *(arr + i + j);
            }
        }

        printf("%d ", max);
    }
}

int main() {
    int n, k;
    scanf("%d", &n);

    if (n <= 0) {
        printf("-1");
        return 0;
    }

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &k);

    window_maximum(arr, n, k);

    return 0;
}
```

>OUTPUT;

Terminal Emulator
Use the command line

Session Actions Edit View Help

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
┌──(priyadarshna㉿kali)-[~]
└─$ gcc day56.c -o day56

┌──(priyadarshna㉿kali)-[~]
└─$ ./day56
8
12 -1 -7 8 -15 30 16 28
3
12 8 8 30 30 30

┌──(priyadarshna㉿kali)-[~]
└─$ ./day56
6
1 2 -3 4 5 -6
2
2 2 4 5 5

┌──(priyadarshna㉿kali)-[~]
└─$ ./day56
3
1 2 3
4
-1

┌──(priyadarshna㉿kali)-[~]
└─$
```